

NORTH-WEST UNIVERSITY (MAFIKENG CAMPUS)

CMPG 325 – COMPUTER NETWORKS

INDIVIDUAL ASSIGNMENT

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CLIENT ID: CLI – 064

**Organisation: Motswaledi Civil And Structural
Engineers (Mafikeng)**

Industry: Engineering

Milestone 1 Document

1. Client Requirements Analysis

Client: Motswaledi Civil & Structural Engineers (Mahikeng) — an engineering consultancy.

1.1 Business context

Motswaledi is a civil and structural engineering consultancy. Day-to-day work centres on structural design calculations, civil site design, and the production of technical drawings (CAD/drafting), supported by a survey/field team that spends time away from the office.

1.2 Departments and network needs

Department	Function	Traffic profile
Management/Admin	Firm leadership, HR, finance	Light, sensitive (needs isolation from general staff)
Structural Engineering	Design calcs, structural CAD	Moderate-heavy, large file transfers
Civil Engineering	Site/civil design work	Moderate-heavy
Drafting & CAD	Drawing production, plotting	Heavy (large drawing files, plotters)
Site Survey / Field	Surveyors, often mobile	Wireless access needed
Reception	Client-facing, visitor point	Light
Servers	File server (drawings), print server	Central, needs highest protection
Legacy Devices	Old plotter/scanner/terminal without modern security	Must be isolated per the brief's constraint

1.3 Design constraint

Legacy devices without modern security features must still be accommodated safely. This is addressed by placing them in a dedicated VLAN (80) with ACLs on the core switch restricting them to only the traffic they need (e.g. printing to the file server), rather than open access to the rest of the network.

1.4 Change request (CR1)

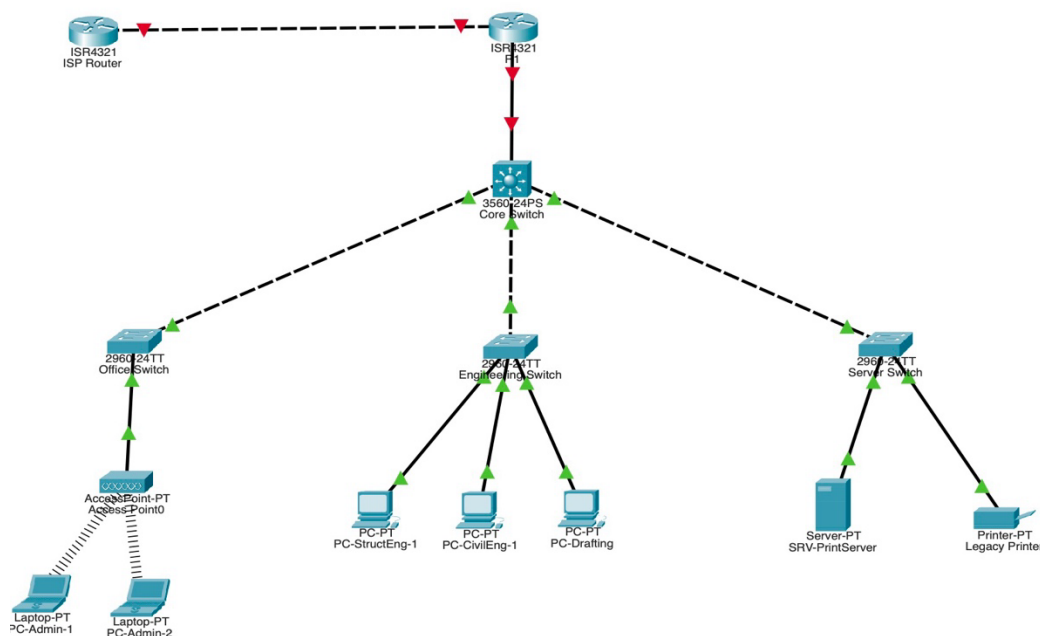
The client will sign 8 additional staff into one department, and the network must absorb this without a redesign. Structural Engineering was chosen to receive the growth allowance, since it is the largest technical team — its VLAN (20) is sized at a /26 (62 usable addresses) from the outset, well beyond current headcount.

1.5 Assigned technical challenge

Default Routing (edge/ISP path design) — Intermediate difficulty. The edge router (R1) uses a single static default route out to the simulated ISP for all internet-bound traffic; the ISP router holds a static route back to the full 172.30.40.0/23 block via R1's WAN interface. No dynamic routing protocol is required or used for this design

2. Physical Topology

The physical hierarchy places a single edge router (R1) between the ISP and the internal network, with a Layer-3 core switch handling inter-VLAN routing internally, and three access switches serving grouped departments

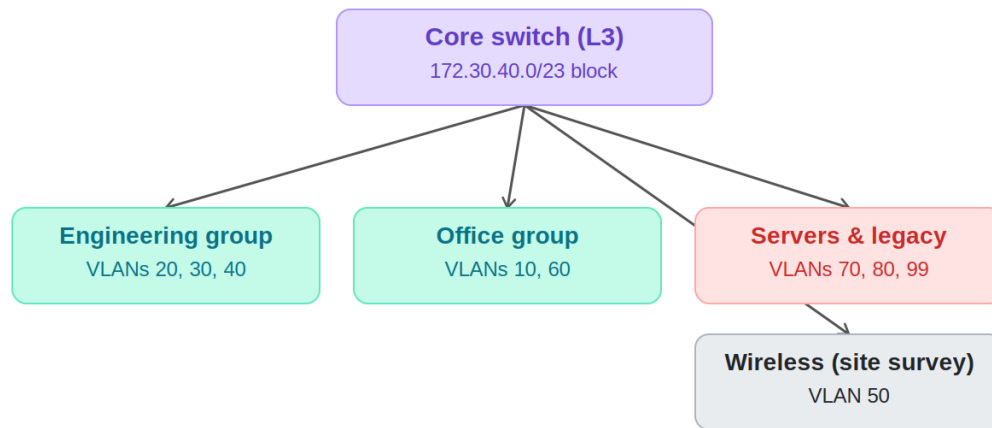


2.1 Design rationale

- single edge router (R1) gives one clean place to enforce the assigned Default Routing challenge — internal subnets are already known via direct connections, so R1 only needs one static default route out.
- The Layer-3 core switch handles all inter-VLAN (east-west) traffic, keeping it off the edge router entirely.
- Three access switches group departments by function: Office (VLANs 10, 60), Engineering (VLANs 20, 30, 40), and Server/Legacy (VLANs 70, 80).

A wireless access point off the Office switch carries VLAN 50 for the Site Survey / field team

3. Logical Topology



VLAN	Name	Group
10	Management / Admin	Office
20	Structural Engineering	Engineering
30	Civil Engineering	Engineering
40	Drafting & CAD	Engineering
50	Site Survey (Wireless)	Wireless
60	Reception	Office
70	Servers	Servers & Legacy
80	Legacy Devices	Servers & Legacy
99	Native / Switch Management	Servers & Legacy

4. IP Addressing Plan (VLSM on 172.30.40.0/23)

Your block spans 172.30.40.0 – 172.30.41.255 (512 addresses). Here's the carve-up:

VLAN	Name	Subnet	Range	Usable Hosts	Gateway
20	Structural Engineering	172.30.40.0/26	.0–.63	62 (room for +8 growth)	172.30.40.1
30	Civil Engineering	172.30.40.64/27	.64–.95	30	172.30.40.65
40	Drafting & CAD	172.30.40.96/27	.96–.127	30	172.30.40.97
10	Management/Admin	172.30.40.128/28	.128–.143	14	172.30.40.129
60	Reception	172.30.40.144/28	.144–.159	14	172.30.40.145
50	Site Survey (Wireless)	172.30.40.160/27	.160–.191	30	172.30.40.161
70	Servers	172.30.40.192/29	.192–.199	6	172.30.40.193
80	Legacy Devices	172.30.40.200/29	.200–.207	6	172.30.40.201
99	Native/Mgmt (switches)	172.30.40.208/29	.208–.215	6	172.30.40.209
—	WAN link (Edge↔ISP)	172.30.40.216/30	.216–.219	2	—

That leaves 172.30.40.220 – 172.30.41.255 completely untouched — a large reserve block for any future department growth beyond CR1.

Why VLAN 20 is sized at /26: Structural Engineering is the natural pick to absorb "8 additional staff in one department" since it's your largest technical team. Sizing it at 62 usable addresses from the start means when the change request lands, you just plug new devices into the existing VLAN — no re-addressing, no redesign, which is exactly what the rubric wants to see justified.

5. Github

<https://github.com/mazibukothabanggift/CMPG-325-Individual-Semester-Project.git>